Matrix-vector products

Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathbb{R}^{m\times n}$ and $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$, interpreted as a column vector.

Column definition: $A\mathbf{x} = \sum_{j=1}^{n} x_j \mathbf{a}_j \in \mathbb{R}^m$. The input length must equal the number of columns of A.

Row dot-product theorem: for $1 \le i \le m$, $(A\mathbf{x})_i = \text{row}_i(A) \cdot \mathbf{x} = \sum_{j=1}^{n} a_{ij}x_j$.

Product properties: for $A, B \in \mathbb{R}^{m\times n}$, $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $r \in \mathbb{R}$,

$$A(\mathbf{x}+\mathbf{y}) = A\mathbf{x} + A\mathbf{y}, \qquad (A+B)\mathbf{x} = A\mathbf{x} + B\mathbf{x},$$

$$A(r\mathbf{x}) = r(A\mathbf{x}) = (rA)\mathbf{x}.$$

Standard basis and geometry

Standard basis vector: $\mathbf{e}_j \in \mathbb{R}^n$ has entry 1 in position j and 0 elsewhere, for $1 \le j \le n$.

Basis expansion and column images: $\mathbf{x} = \sum_{j=1}^{n} x_j \mathbf{e}_j$ and $A\mathbf{e}_j = \mathbf{a}_j$.

Unit-square visualization: for $A \in \mathbb{R}^{2\times 2}$, the square's corner images are $\mathbf{0}, \mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_1 + \mathbf{a}_2$. They form a parallelogram, possibly collapsed to a segment or a point.

Linear maps

Function: $F : \mathbb{R}^n \to \mathbb{R}^m$ assigns one output to each input. Its **domain** is $\mathbb{R}^n$; its **co-domain** is the specified output space $\mathbb{R}^m$.

Matrix map: for $A \in \mathbb{R}^{m\times n}$, $T_A : \mathbb{R}^n \to \mathbb{R}^m$, $T_A(\mathbf{x}) = A\mathbf{x}$.

Linear map / linear transformation: $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear if, for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and $c \in \mathbb{R}$,

$$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \qquad T(c\mathbf{v}) = cT(\mathbf{v}).$$

Preservation of linear combinations: for linear T, any $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and $c_1, \dots, c_k \in \mathbb{R}$,

$$T\left(\sum_{j=1}^{k} c_j \mathbf{v}_j\right) = \sum_{j=1}^{k} c_j T(\mathbf{v}_j).$$

Matrix representation theorem: every matrix map is linear; every linear $T : \mathbb{R}^n \to \mathbb{R}^m$ has the unique matrix

$$A = [T(\mathbf{e}_1) \ \cdots \ T(\mathbf{e}_n)] \in \mathbb{R}^{m\times n}, \qquad T(\mathbf{x}) = A\mathbf{x}.$$

Zero-vector test: linearity implies $T(\mathbf{0}) = \mathbf{0}$; this condition alone does not prove linearity.

Affine maps

Affine map: $F : \mathbb{R}^n \to \mathbb{R}^m$, $F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, where $A \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$ is fixed. A linear map is followed by a **translation**, adding $\mathbf{b}$ to the output.

Linear exactly when: $\mathbf{b} = \mathbf{0}$, since $F(\mathbf{0}) = \mathbf{b}$.